OPSC6312

POE Part 1 - Planning

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Introduction

In this report I will go over the planning for the app I have been tasked to make. From the logo to the UI to the database, all will be described to the best of my ability.

App overview

Summary of app

It will essentially be an app that you can open, take a picture of an insect you see nearby, and post the insects picture and its name/species which will then show as a pin on a real time map to let others know of the bugs location. I imagine it as Pokémon Go but for autistic people (as if Pokémon isn't), where you can open the app and see if there are any cool bugs near you to check out. This would be especially cool for rare or migratory insects, as these are often already being documented for research. It would also be useful for monitoring invasive bugs. Mostly it would just be to satisfy the childhood urge many of us still harbour to yell "Yo check out this weird bug!" to all around so you can all gawk at the cool bug together. Other users would be able to interact with the sighting by possibly leaving a reaction of sorts that means "Yup, I saw it too" to make it more interesting and communal. Sightings would probably be taken down after an hour or so as bugs tend to move around, possibly I could include a timed progress bar to let users know how likely it is that the bug is still there that can reset or increase based on the number of "Yup, I saw it" reactions it gets and how recently it got them. If I had the funding I would include an identifying AI or something to identify the bugs as you post, but I do not possess such riches. Perhaps if the Picture Insect developers are willing to hear me out, I could work with them to integrate my little project into their existing apps ecosystem.

App name

Iterations

- Bug spotting
- Look! A bug!
- Insect Go
- Pin-a-Bug

Final Choice

- Pin-a-Bug

Rough icon concept



Figure 1 App Logo concept

(Figma, 2025)

Innovative features

As far as I understand, "innovative" usually means "has never been done before and is a net positive". The features in the app will hopefully be a net positive, but I don't think anything that I will be adding fits the "never been done" aspect. There is no real innovation, the idea has been done before just with a different coat of paint. Maybe I am misunderstanding this projects definition of "innovative" but I do not think I am "innovating" in this app. That is not a bad thing either, sometimes trying to be new and different for the sake of being called innovative can make a worse product. I have seen what works, and if it ain't broke, don't fix it.

App requirements

Requirements as per POE instructions

- User must register through the app and set a username and password
- User must log in to use the app using registered username and password
- Username and password must be stored and encrypted in some way
- App must use SSO in some way *
- The app must contain configurable settings
- Users must be able to alter these settings
- Make use of REST API
- API must be connected to a database
- App must be able to perform some form of offline activities
- Some form of synchronization must occur when internet connection is restored *
- Include some form of Real-time notification system *
- App UI must be available in at least 1 other language besides English *

* Only expected to be presentable in the final POE submission, as such these requirements might not appear fully prototyped in this document or in the Part 2 content

App specific requirements

- Must allow for image uploading
- Must be able to display accurate live map of users location
- Track which user uploaded which image in case of inappropriate behaviour
- Do not show identifiable information of poster to protect from stalking
- Contain settings to change UI colours (app will have set themes to choose between)
- Accurate location service
- Include interface to interact with other users posts
    - Button to confirm that the insect is at that location
    - Button to deny that the insect is at that location
- Personal log of all bugs reported by user that user can see
- Ensure that all pins are removed from the map after 1 hour of no interaction
- Report button to report inappropriate images which will hide the image and pin from the user who reported it
- User must agree to and confirm that they agree to the rules of the app before being permitted to make an account and again before they create their first post
- Some way to deter users from doxing themselves by posting exact locations when it's a persons house since the pins should be almost exactly accurate to where the image was taken and this may lead to privacy issues

Experimental features

These are features that I would like to implement but I do not guarantee that they will be implemented. This could be due to scope creep, integration issues, user feedback and so forth. **Do not hold me to these features**.

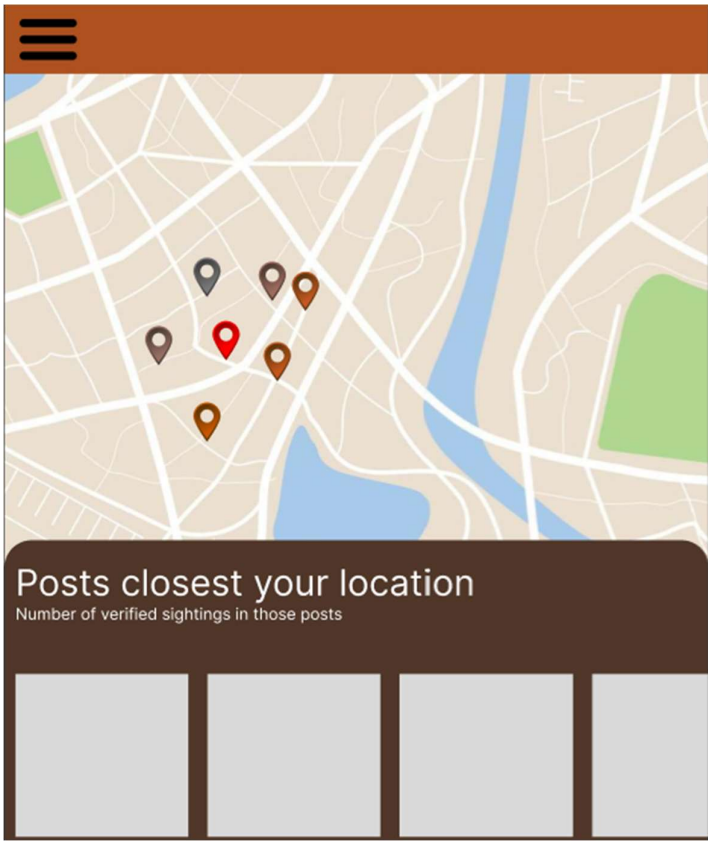
- Interface for user to give name of bug when uploading the image
- Interface for other users to correct/add the name of the bug seen in others posts
- Show "hotspots", places that get the most pins daily
- Ranking system for users to see if a post is from a very active user or very inactive user
- Reported users who are found guilty of inappropriate behaviour are offered a one time grace where they must type out the rules one by one to be unbanned. This will ensure that they will have no defence if they incur another infraction
- Feature to try prevent people being weird about addresses and not dox people. Mayhaps a pop up to ask the user "Hey is this your/somebody's literal house?" which when answered "yes" will inform the user that the location (depending on what the user chooses) will either:
    - Be redacted and the post will not be posted publicly but the sighting will still be recorded for data purposes
    - Be approximated by a set range (block, neighbourhood, suburb, etc) as an off-centre circle or an outline of said range based on the map borders
    - Be uploaded exactly as it would for a public area
- Detect when user wants to post about a location that has recently got a post and ask them if this is the same bug that that user posted about so they can be grouped together on the map.

Unobtainable features

This is a list of features that I am sure I cannot deliver on given my current situation. Whether it be financial or other, I am incapable of implementing these features, full stop. They would, however, exist if the app were made by a professional team with money and such.

- Image identification to auto identify the bug
- Image scanning to prevent inappropriate images from being posted
- Experts on call to verify the identification of the bug and correct it in real time
- Ability to collect data about what bugs are most common where, at what time of day and year and what their morphology is

UI mock-ups



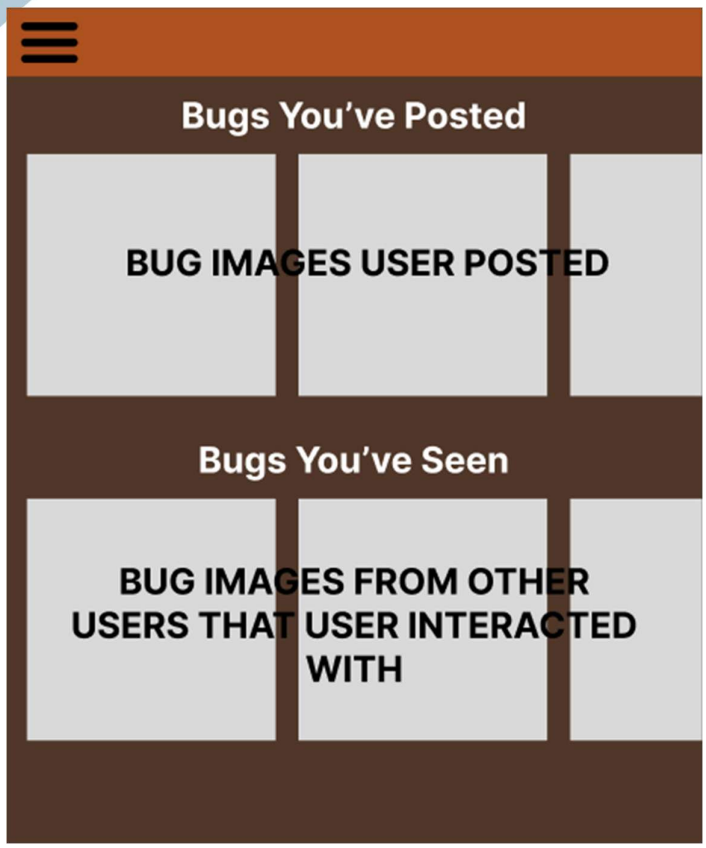
This is the main screen that the user will see when opening the app. They can see where they are (red pin) and can see recent bug sightings (brown pins) around them. The pins will have different saturations depending on recency of post and recency of interactions with that post. The bugs nearest you will be in a quick access bar at the bottom of the screen for user convenience. The latest and most verified posts will be first in line. (Figma, 2025)

Figure 2 App home screen



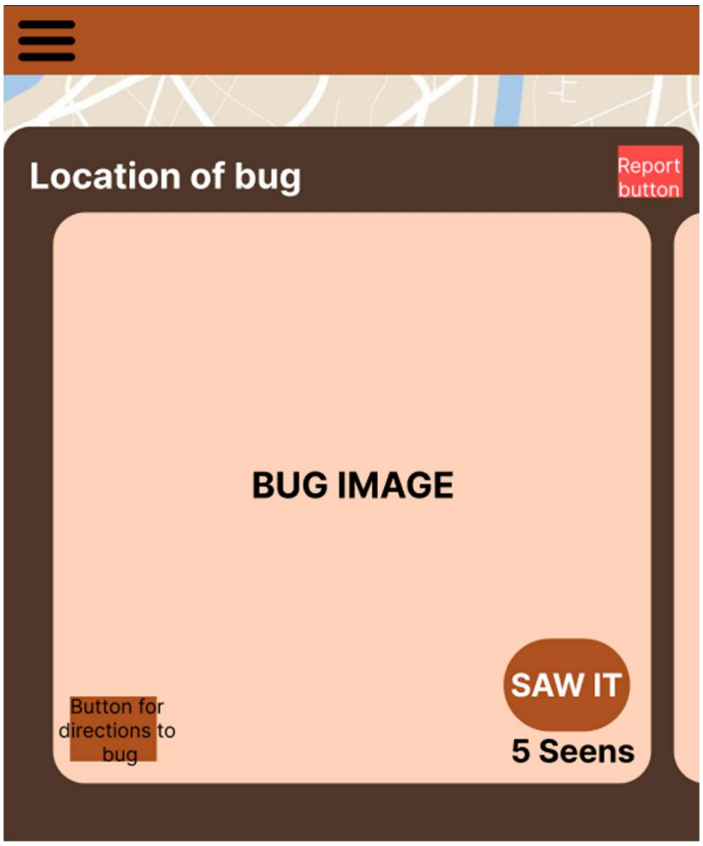
This is the make a post screen where the user choses the location, adds the image, suggests a bug name and can hit post. (Figma, 2025)

Figure 3 App new post screen



This is where users can see the posts they've made and the posts they've interacted with before. Might make the users posts always available for them to look at even when offline. (Figma, 2025)

Figure 4 App post history screen



This is what you'd see when you tap a pin or select a post from the quick access bar. It will show the bug image, button to say you went and saw it, how many people have seen it, button to direct you to those specific coordinates and button to report the post for being inappropriate. If there are other posts that are made about the same bug in the same coords then you will be able to swipe to see that post too. (Figma, 2025)

Figure 5 App bug posts pop up

Site map

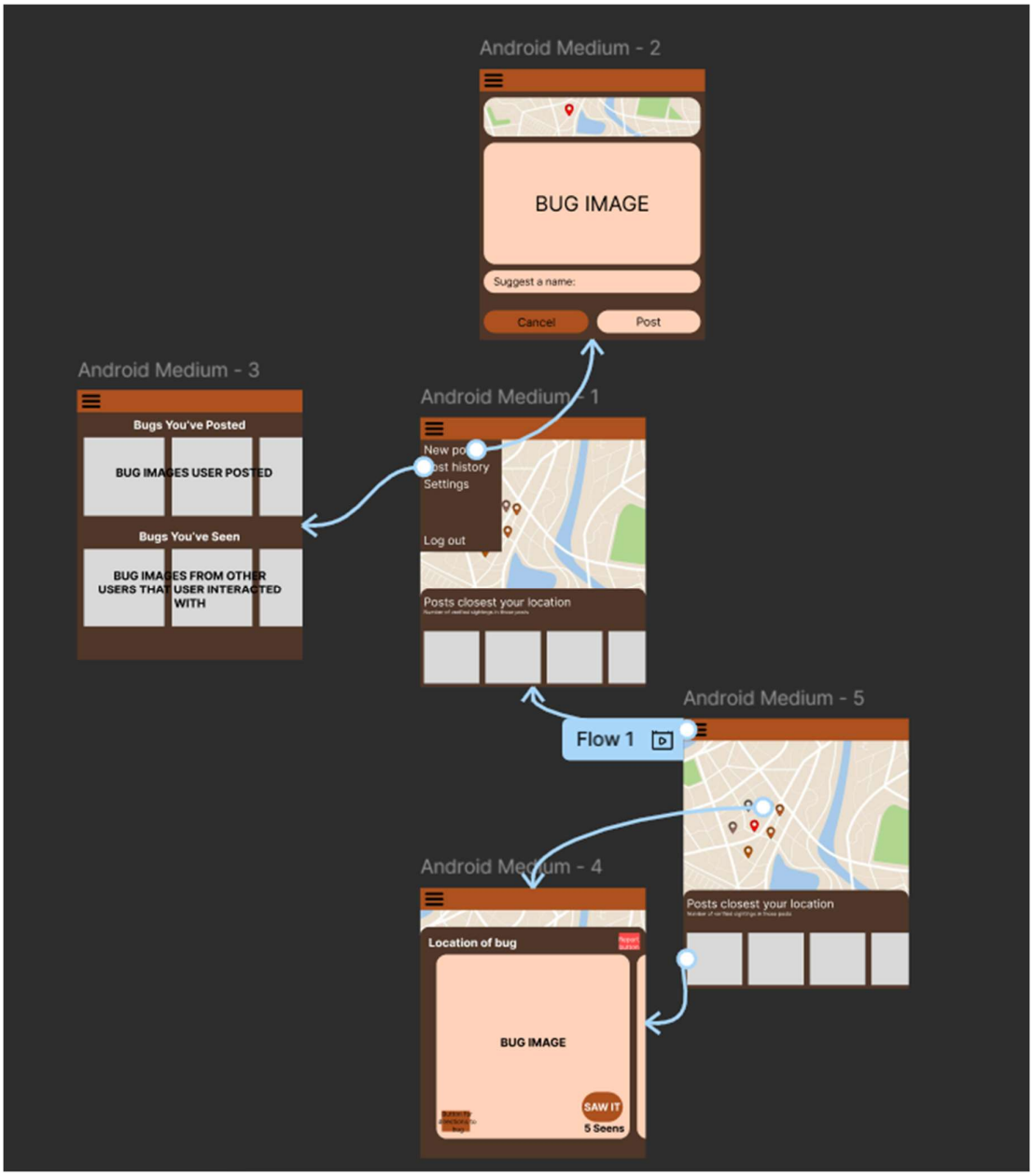


Figure 6 App Site map

(Figma, 2025)

Disclaimer: As the SDLC goes on, especially with Agile development, designs will be relooked at and redone as one comes across new problems or new ideas while in the other phases of the SDLC. The final app may not look exactly like the initial designs, but that is to be expected with both big experienced developers and inexperienced college students.

API design

When beginning this project I thought I could just use the Google maps SDK for maps. I have since reread the instructions and I see it must specifically be a REST API, which I am not sure the Google one is.

I think I may use GeoAppify as it has the pin feature I am looking for. I will need to get outside help to decide which REST API would be best for the application I want to make. I will most likely figure out what API I need and how it works by tinkering with it in part 2, I learn best by doing. I can also reach out to people in the field who know about APIs and especially map APIs to help me out. The Picture team would be an ideal resource given that they already have a working map with features that I want, but they are understandably not able to help out on the matter.

(Akella, 2025, OpenStreetMap, n.d., Documentation - Leaflet - a JavaScript library for interactive maps, n.d., Maps, APIs and components | Geoapify Location Platform, 2018, Geocoding API | Search addresses and locations | Geoapify, 2020,)

UML Diagram

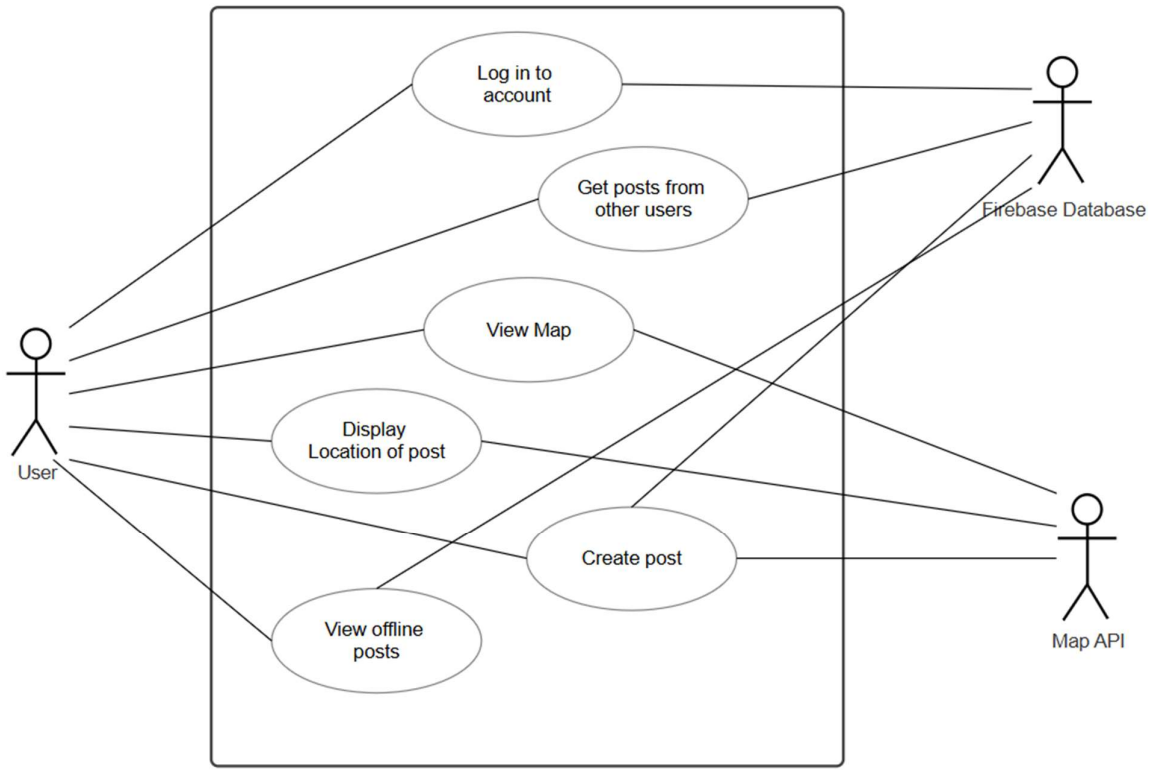


Figure 7 UML Diagram

(Free UML Diagram Tool - Free Templates Make UML Design Easy, n.d.)

Capturing user data

It will need to store (on Firebase):

- User ID of sorts (varchar as it will probably be a mix of numbers and letters)
- User password (varchar for different password styles)
- Each posts unique ID (also probably varchar)
- Location of post (the geo datatype for coordinates)
- Images uploaded (unique datatype?)
- IDs of posts the user has interacted with (same datatype as post IDs)

It may also need to store the following locally on the users device (for offline):

- Copies of images the user uploaded (store in a folder and have database file store the location of the pic on device?)
- The settings states (what theme, language, etc)

Project plan

See the Gantt chart pdf submitted alongside this document.

The part 3 finish date is a guess, I don't know when part 3 will be due.

Conclusion

For the app I have thought about each aspect of the experience. I have researched and brainstormed both functional and non functional requirements, and have documented my findings and thoughts as best I can.

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